Sanskar Jain

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Summary

Backend engineer with 2+ years of experience building scalable distributed services, workflow engines, and data-heavy platforms at Verisk Analytics. Designs state machines, async pipelines, and fault-tolerant architectures handling 200K+ weekly requests in high-scale production. Thinks in systems, data models, and trade-offs; owns architecture end-to-end from design to production reliability.

Skills

Languages: C#, Java, Python, TypeScript
Backend & Systems: .NET 8, ASP.NET Core, Spring Boot, Microservices, REST APIs, State Machines, Workflow Engines
Distributed Systems: Kafka, AWS SQS/SNS, Redis, Event-Driven Architecture, Idempotency, Retry/Backoff
Cloud & DevOps: AWS (S3, SNS, SQS, EKS, ECS, Lambda), Docker, Kubernetes, Terraform, CI/CD
Databases: MongoDB, PostgreSQL, MySQL
Observability & Tools: Splunk, Structured Logging, Azure DevOps, Swagger, Postman

Experience

Verisk Analytics Jan 2024 – Present
Software Engineer II, Hyderabad, India

- Designed and built **RulesRunner**, a workflow engine microservice using **SQS** and state-machine patterns that orchestrates compliance checks across multiple stages, reducing client-side latency from **3000ms to 500ms** (**83%**) and enabling flexible scaling across workflows.
- Architected a scalable data-heavy backend for **Xactimate's** import/export of **30+ data types** using async processing, S3 presigned URLs, and MongoDB — reducing data transfer time by **38%** and user-reported issues by **25%**.
- Eliminated message loss under failure conditions by designing **AWS SNS/SQS/Lambda** pipelines with idempotent processing, dead-letter queues, and exponential retry/backoff for reliable distributed delivery.
- Owned production architecture for APIs handling **200K+ weekly requests**, implementing structured error handling, input validation, circuit breakers, and **Swagger**-documented contracts.
- Improved API response time by **34%** via **Redis**/ElastiCache caching, connection pooling, and targeted query optimisation on hot-path endpoints serving high-scale traffic.
- Cut technical debt by **55%** and improved runtime performance by **30%** through migrating legacy services to **.NET 8**, modernising dependency injection, async patterns, and SDK tooling.
- Reduced mean time to diagnose production incidents by instrumenting services with structured logging and **Splunk** dashboards, surfacing error trends and latency spikes in real time.
- Designed data models in **MongoDB** and **PostgreSQL** optimised for high-throughput read/write patterns, balancing consistency trade-offs across distributed services.
- Led architecture decisions — evaluated trade-offs between synchronous vs. event-driven patterns, choosing SQS fan-out for decoupled scaling and resilience under peak load.

Projects

Expense Tracker – Distributed Microservices Platform Aug 2023 – Dec 2023

- Built a distributed platform using **Spring Boot** with **5+ microservices**, async **Kafka** event-driven communication, and stateful workflow coordination across services.
- Integrated **Kong API Gateway** for routing, load balancing, and authentication; simulated **1000+** concurrent transactions with failure recovery.
- Implemented distributed state management with eventual consistency, handling partial failures and compensating transactions across service boundaries.

Certifications

AWS Certified Cloud Practitioner Amazon Web Services — Oct 2024

Education

Bachelor of Technology in Computer Engineering
Vishwakarma Institute of Technology, Pune (S.P.P.U) 2024
CGPA: 8.8